Closing the loop with AlphaFold2

Cyclic peptide structure prediction and design using AlphaFold2

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AfCycDesign: reuse a protein model's learned geometry by telling it that the peptide is a ring.

Can a linear-protein model design a small peptide ring?

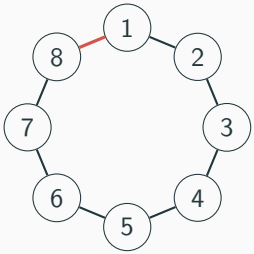
Cyclic peptides offer compact, constrained scaffolds for binding protein surfaces. Their structural data are too sparse to readily train a dedicated model from scratch.

The transfer hypothesis
Turns and local interactions in macrocycles also occur in larger proteins. AlphaFold2 may already know much of the required geometry, but its input must describe the correct chain topology.

The paper tests a ladder of capabilities: native structure prediction $\rightarrow$ fixed-backbone sequence design $\rightarrow$ new scaffolds $\rightarrow$ binders.

Introduction; Results. Scope: head-to-tail cyclic peptides, predominantly canonical L-amino acids.

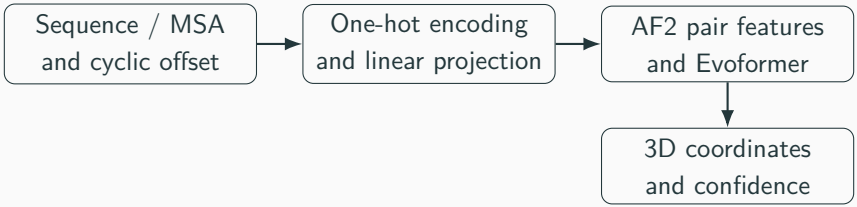
A concrete example: residue 8 is next to residue 1



For a linear 8-mer, terminal residues have sequence separation 7. In the ring they are adjacent: separation magnitude 1. The cyclic offset changes the relative positional features for residue pairs. Signs retain chain directionality; long-range conventions have alternative implementations. Circularly relabeling the same sequence should describe the same molecule.

Fig. 1a; Supplementary Figs. 1, 4. Diagram is an original conceptual example, not a reproduction.

What changes inside the prediction pipeline?



The pretrained network weights are reused. The key intervention is the residue-pair topology supplied to the model. For a target–peptide complex, apply the cyclic offset to the peptide chain; leave the protein target's encoding linear.

Results: Structure prediction; Design and validation of binders. No macrocycle-specific retraining is reported.

Native structures: what exactly is the benchmark?

Inputs and reference

- 80 PDB NMR structures.
- Canonical amino acids; lengths below 40 residues.
- Diverse folds, including disulfide-rich peptides.

Evaluation protocol

- MSA, 6 recycles, all five AF2 models.
- Select model with highest pLDDT.
- Report its lowest backbone-heavy-atom RMSD to members of the NMR ensemble.

pLDDT is shown on a 0–1 scale here. RMSD measures structural agreement in Å. Neither metric directly measures binding or stability.

Fig. 1; Methods: Structure prediction. Authors state these structures were excluded from AF2 training; homolog-level independence is a separate issue.

Prediction works often; confidence is useful but imperfect

Readout	Result
Median pLDDT / backbone-heavy-atom RMSD	0.92 / 0.8 Å
Accurate and confident: RMSD < 1.5 Å, pLDDT > 0.7	58/80
Accuracy among high-confidence predictions (> 0.85)	44/55 (80%)
Single-sequence accurate/confident predictions	49/80

No disulfide connectivity was explicitly supplied. Correct connectivity emerged in most high-confidence successful predictions. A high-confidence prediction is an enriched candidate, not a guarantee.

Fig. 1b–f; Results. The 58/80 and 49/80 runs use different sequence-information settings.

The ablations locate the useful information

Single-sequence, single-recycle offset test	Successes / 80
Type 1: alternative long-range sign convention	52
Type 2: accurate cyclic positional labeling	49
Type 3: only adjacent pairs retain local offsets	48

Removing cyclic offsets worsens predictions. Preserving ring adjacency captures much of the gain in this benchmark; authors recommend Type 2 as the default. Seven cases had an alternative AF2 model within 1.5 Å when the top-pLDDT model missed that threshold: inspect all five models downstream.

Fig. 1; Supplementary Figs. 2–4. Design sections used Type 1; do not conflate ablation settings with the main benchmark.

Sequence design: optimize the input, hold the network fixed

For a desired backbone, turn its pairwise distances into a target distogram: a binned distance distribution for every residue pair.

$$\mathcal{L}_{\text{fixbb}} = \text{CCE}(D_{\text{target}}, D_{\text{AF2}}(S))$$

S is an $N \times 20$ sequence representation. Gradients update sequence logits so AF2's predicted distances match the target backbone.

Why a continuous sequence?
A discrete amino-acid choice is not differentiable. Optimize continuous logits, move toward softmax probabilities, then a hard one-hot sequence with a straight-through estimator.

Supplementary Fig. 6; Methods: Cyclic peptide design. CCE = categorical cross-entropy; network weights are not the design variables.

A redesign case connects the score to a real structure

A Rosetta-generated 13-mer backbone with a short helix had a sequence with a small predicted energy gap to alternative conformations.

	Rosetta sequence	AF2 redesign
Name	RRR13.1	RAR13.1
Rosetta energy gap	< 2 kcal/mol	~ 6 kcal/mol
Mutations relative to starting sequence	—	12/13

Racemic X-ray crystallography validates RAR13.1: Cα RMSD 0.3 Å to design; 10/13 sidechain rotamers match.

Fig. 2a–c. Energy gap is a computational folding proxy; crystallography is the experimental structural test.

Large-scale redesign: a gain in the optimized metric

Same 3274 cyclic 13-mer backbones	pLDDT > 0.9
Rosetta sequence design	63
AfCycDesign sequence design	1145

AfCycDesign sequences use more hydrophobic residues and prolines. **Interpretation:** AF2 optimization finds sequences AF2 prefers. The matched-backbone comparison is useful, but pLDDT is not an independent validation endpoint. Rosetta was restricted to the same 20 canonical amino acids for this comparison; it did not use its full heterochiral design capability.

Fig. 2d,e; Results: Sequence redesign. One redesign is structurally validated, not all 1145 high-confidence clusters.

Hallucination removes the target backbone

Now both sequence and predicted structure can change. The search rewards confident, compact cyclic structures:

$$\mathcal{L}_{\text{hallucination}} = (1 - \text{pLDDT}) + \text{PAE}/31 + \text{contact loss}/2.$$

PAE is predicted aligned error; the contact term promotes intramolecular interactions. Random sequence initialization and a brief linear-offset phase precede cyclic optimization. The cyclic hallucination schedule uses 50 soft/logit, 50 temperature-annealing, and 10 hard-sequence steps. Cluster by backbone torsion-bin strings, grouping circular permutations together. Then filter high-confidence clusters and check selected sequences with Rosetta energy landscapes.

Methods: Cyclic peptide design; Figs. 3–4. General fixbb protocol has a longer 300/200/10 schedule.

A large predicted scaffold library, not 10,681 solved structures

Length	Designs sampled	Unique torsion clusters	Clusters pLDDT > 0.9
7	48,000	10,009	200
8	48,000	13,210	296
9	48,000	19,746	486
10	48,000	22,238	1352
11	48,000	28,553	1846
12	48,000	27,643	2861
13	48,000	27,330	3825
Total	336,000	148,729	10,866

The paper reports a released high-confidence set of **10,681**. Its per-length counts sum to 10,866; these are kept distinct here rather than silently reconciled.

Figs. 3–4 and main-text counts; Supplementary Fig. 7: cluster discovery has not saturated.

Crystallography tests selected hallucinated designs

Design	Cα RMSD (Å)
RH7.1	0.9
RH8.1	1.0
RH9.1	0.5
RH10.1	0.3
RH11.1	0.3
RH12.1	0.4
RH13.1	0.8

One design per length was selected using pLDDT > 0.9 *and* Rosetta $P_{\text{near}} > 0.9$. Small designs use turns and internal hydrogen bonds; larger examples contain short helices or sheets without extra disulfide crosslinks. Strong evidence of achievable atomic accuracy. This selected panel does not estimate the success rate of all generated sequences.

Figs. 3–4; Supplementary Fig. 13. Rounded RH8.1 value is 1.0 Å; the abstract broadly describes all eight monomers as below 1.0 Å.

Two checks that temper a confidence-only story

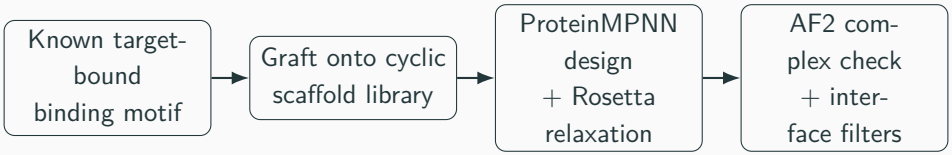
A second sequence designer finds complementary solutions. On 22,182 ten-residue backbones, 608 hallucinated sequences and 505 ProteinMPNN sequences re-predict with pLDDT > 0.9; only 157 backbones overlap. **Mutational tolerance is assessed**

computationally. For the 10,681-scaffold set, 98% of single substitutions do not reduce pLDDT by 0.2 or more. Glycine and proline changes are common exceptions. Neither

comparison establishes experimental folding robustness. It suggests complementary search strategies and candidate sites for optimization.

Supplementary Figs. 8–9; Results. Rosetta P_{near} is an additional computational check, not measured stability.

From a stable scaffold to a binder: preserve a known motif



For MDM2, graft p53's five-residue FSDLW motif; keep phenylalanine and tryptophan fixed while redesigning surrounding residues. The expanded 7–16-residue library contains 24,104 high-confidence scaffolds. The binding-site geometry comes from a known complex.

Fig. 5a; Methods: MDM2 peptide binder design. This demonstration is motif-guided, not target-only de novo interface discovery.

MDM2: track the funnel and the experimental denominator

Stage	Count
Grafted cyclic peptides redesigned	18,722
Computationally filtered / manually selected	27 / 16
Successfully synthesized / assayed	14
More than 50% inhibition at 50 μM	5
Best three followed with dose response	3

RMG_14, RMG_15, RMG_16: $IC_{50} = 0.3384$, 1.383, 0.8702 μM. The isolated grafted motif is inactive at 50 μM. AlphaLISA measures competition with an MDM2 ligand, in triplicate. The 5/14 hit fraction describes the heavily selected tested set.

Fig. 5b; Supplementary Fig. 10; Results. Interface confidence and Rosetta metrics precede manual selection.

MDM2: structure and affinity provide different evidence

Binding mode: the RMG_14–MDM2 complex is solved at 1.7 Å resolution. Peptide Cα RMSD is 1.0 Å after aligning MDM2, or 0.8 Å after aligning the peptide alone. **Affinity:** SPR gives $K_D = 3.0$ μM for RMG_14. Five substitutions, including D- and noncanonical residues, improve K_D to 294 nM for RMG_14c. The IC_{50} of 338.4 nM and the K_D of 3.0 μM are different assay endpoints. They should not be treated as interchangeable affinity estimates.

Fig. 5c,d; Supplementary Figs. 11, 14. Noncanonical optimization uses external chemical/physics-based reasoning beyond AF2's alphabet.

Keap1 extends the demonstration beyond a helical motif

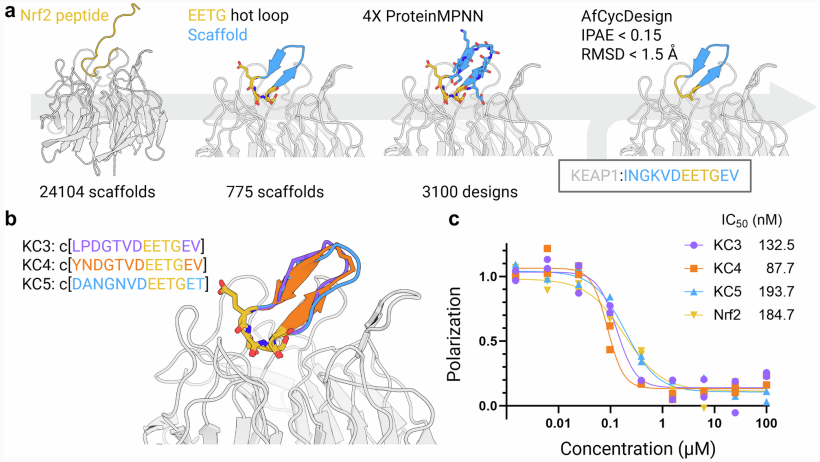
Nrf2's EETG loop motif matches 775 scaffolds. Four sequence-design rounds yield 3100 candidates; six are selected, three synthesized successfully. Competitive fluorescence polarization, two replicates per concentration:

Peptide	IC_{50} (nM)
KC3	132.5
KC4	87.7
KC5	193.7
Nrf2 linear peptide	184.7

All three tested cyclic peptides inhibit the interaction below 200 nM; KC4 is about twice as potent as the linear comparator. 798/1014 known hot loops can be geometrically grafted onto at least one scaffold. This is computational coverage, not 798 validated binders. No experimental Keap1–designed-peptide structure is reported.

Fig. 6; Supplementary Fig. 12; Methods: Keap1 assay. EETG follows pipeline/figure; one results sentence inconsistently says DEET.

Read the original Keap1 evidence chain



Unmodified Fig. 6, Rettie et al., Nat Commun 16:4730 (2025). CC BY-NC-ND 4.0. Panel b contains design models; panel c contains experimental competition data.

What has been demonstrated, and what remains open?

Supported by experiments

- Eight selected monomer structures closely match designs.
- A designed MDM2 complex matches the intended binding mode.
- Selected MDM2 and Keap1 peptides compete in biochemical assays.

Still unresolved

- Solution conformational ensembles; crystals may select one state.
- Protease/serum stability, permeability, cellular activity and selectivity.
- Generalization to arbitrary interfaces and noncanonical chemistry.

Synthesis failures and confidence-based selection matter when interpreting the library's practical yield. Confidence optimization creates a need for independent experimental filtering.

Discussion; Methods and experimental denominators. These limitations constrain therapeutic interpretation, not the demonstrated structural accuracy.

Discussion: what would make the next experiment decisive?

1. Randomly sample across confidence and P_{near} bins; measure synthesis success and solution structures. Does confidence calibrate to experimental yield?
2. Compare grafted cyclic peptides with matched linear and motif-only controls. How much gain comes from cyclization versus new contacts?
3. Test a new target with a predeclared selection protocol, then measure selectivity and cellular activity. How portable is the scaffold strategy?

Take-home: a small change in topology encoding unlocks useful transfer from AF2 to macrocycles. The strongest evidence is selected atomic structure validation; the broad library remains a computational resource.

Discussion prompts are our interpretation. Paper: doi.org/10.1038/s41467-025-59940-7. Code: ColabDesign, doi:10.5281/zenodo.13309081; scaffolds: doi:10.5281/zenodo.15164650.